

Curriculum Vitae and List of Publications

PROF. DR. EMRE ONUR KAHYA

Contact Information

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Education

M.Sc., Computer Science, 2021, **Mila - Quebec AI Institute, Université de Montréal**, Canada
Thesis: Identifying Electrons with Deep Learning Methods

Ph.D., Physics, 2008, **University of Florida**, Gainesville, Florida, USA.
Thesis: Quantum Gravitational Correction to Scalar Field Equations During Inflation

M.Sc., Physics, 2002, **Middle East Technical University**, Ankara, Turkey.
Thesis: Higher dimensional metrics of colliding gravitational plane waves

B.S., Physics, 2000, **Middle East Technical University**, Ankara, Turkey.

Research Interests

Machine Learning, Artificial Intelligence, Cosmology, Alternative Theories of Gravity, Quantum Field Theory

Employment History

Visiting Researcher, Mila - Quebec AI Institute, 2019–2021.

Faculty Member, Istanbul Technical University, 2012–present.

Postdoctoral Researcher, Friedrich Schiller University Jena, 2009–2012.

Postdoctoral Researcher, Koç University, 2008–2009.

Teaching Assistant, University of Florida, 2002–2008.

Teaching Assistant, Middle East Technical University, 2000–2002.

Research Projects

TÜBİTAK 1001, “Carrollian Field Theories”
E. O. Kahya (PI) (2026–2028).

Optimization Methods in Machine Learning,
Project Supported by Higher Education Institutions, E. O. Kahya (PI) (2019–2023)

IVADO Project, Machine learning for the analysis of the Large Hadron Collider Data at CERN,
Yoshua Bengio (PI), Jean-François Arguin (LR), E. O. Kahya (Researcher) (2019–2021)

TÜBİTAK 1001, “Inflation and Supersymmetric Attractors,”
E. O. Kahya (PI), Project No.: 117F102, Awarded Budget: 264,400 TL, (2017–2019).

TÜBİTAK 1001, “Quantum Effects in the Early Universe,”
E. O. Kahya (PI), Project No.: 112T817, Awarded Budget: 148,220 TL (2013–2015).

European Union, Marie Curie International Reintegration Grant,
E. O. Kahya (PI), Project No.: IRG-247803, Awarded Budget: \$ 137,000 (2009–2012).

Awards

Mustafa Parlar Research Incentive Award, 2017.

TÜBİTAK Science Incentive Award, 2016.

Turkish Academy of Sciences Outstanding Young Scientist Award (TÜBA-GEBİP), 2015.

Scientist of the Year Award, Science Heroes Association, 2015.

Most Valuable Referee, “Astroparticle Physics” (2010).

Publications

1. “Stationary solutions in the small c -expansion of GR”, E. Bal, E. Ekiz, E. O. Kahya, U. Zorba, arXiv:2604.23677 (2026).
2. “Do equation of state parametrizations of dark energy faithfully capture the dynamics of the late universe?”, Ö. Akarsu, M. Caruana, K. F. Dialektopoulos, L. A. Escamilla, E. O. Kahya, J. L. Said, arXiv:2604.12987 (2026).
3. “Hints of sign-changing scalar field energy density and a transient acceleration phase at $z \sim 2$ from model-agnostic reconstructions”, Ö. Akarsu, M. Caruana, K. F. Dialektopoulos, L. A. Escamilla, E. O. Kahya, J. L. Said, arXiv:2602.08928 (2026).
4. “The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics” E. Valentino et. al., Physics of the Dark Universe **49**, (2025) 101965.
5. “Quantization of Carrollian Fermions” E. Ekiz, E.O. Kahya, U Zorba, Phys. Rev. **D111** (2025) 105019.
6. “Holographic boundary conformal field theory within Horndeski gravity” Santos et. al., Journal of High Energy Physics **12** (2024) 1-30.
7. “On the Classification and Feature Relevance of Multiband Light Curves,” Fatma Kuzey Edes-Huyal, Zehra Cataltepe and Emre O. Kahya, AJ, **161** (2021) 004.
8. “Superconformal generalizations of auxiliary vector modified polynomial $f(R)$ theories,” S. Boran, E.O. Kahya, N. Ozdemir, M. Ozkan, U. Zorba, JCAP **04** (2020) 005.
9. “Constraints on differential Shapiro delay between neutrinos and photons from IceCube-170922A,” Sibel Boran, Shantanu Desai and E.O. Kahya, Eur. Phys. J. **C79** (2019) 3, 185.
10. “GW170817 Falsifies Dark Matter Emulators,” Sibel Boran, Shantanu Desai, Emre Kahya and Richard Woodard, Phys. Rev. **D97** Rapid Communication (2018) 041501.
11. “Loop Corrections to Primordial Non-Gaussianity,” S. Boran and E.O. Kahya, Phys. Rev. **D97** (2018) 043507.
12. “Galactic Shapiro Delay to the Crab Pulsar and limit on Einstein’s Equivalence Principle Violation,” Shantanu Desai, Emre O. Kahya, Eur. Phys. J. **C78** (2018) 2, 86.
13. “One loop corrected conformally coupled scalar mode equations during inflation,” Sibel Boran, E.O. Kahya and Sohyun Park, Phys. Rev. **D96** (2017) 105003.
14. “Quantum gravity corrections to the conformally coupled scalar self-mass-squared on de Sitter II: kinetic-conformal cross terms,” Sibel Boran, E.O. Kahya and Sohyun Park, Phys. Rev. **D96** (2017) 025001.
15. “Broken Scale Invariance, Alpha-Attractors and Vector Impurity,” O. Akarsu, S. Boran, E.O. Kahya, N. Ozdemir and M. Ozkan, Eur. Phys. J. **C77** (2017) 5, 306.
16. “Constraints on frequency-dependent violations of Shapiro delay from GW150914,” S. Desai and E.O. Kahya, Phys. Lett. **B756** (2016) 265.

17. "The Galactic One-Way Shapiro Delay to PSR B1937+21," S. Desai and E.O. Kahya, Mod. Phys. Lett. **A31** (2016), 1650083
18. "Constructing an inflaton potential by mimicking modified Chaplygin gas," E.O. Kahya, B. Pourhassan and S. Uraz, Phys. Rev. **D92** (2015) 103511.
19. "The universe dominated by the extended Chaplygin gas," E.O. Kahya and B. Pourhassan, Mod. Phys. Lett. **A30** (2015), 1550070
20. "Higher order corrections of the extended Chaplygin gas cosmology with varying G and Λ ," E.O. Kahya, M. Khurshudyan, B. Pourhassan, R. Myrzakulov, A. Pasqua, Eur. Phys. J. **C75** (2015) 2, 43.
21. "Quantum gravity corrections to the conformally coupled scalar self-mass-squared on de Sitter background," Sibel Boran, E.O. Kahya and Sohyun Park, Phys. Rev. **D90** (2014) 124054.
22. "Observational constraints on the extended Chaplygin gas inflation," E.O. Kahya and B. Pourhassan, Astrophysics and Space Science **353**, 677-682 (2014).
23. "Interacting two-component fluid models with varying EoS parameter," M. Khurshudyan, B. Pourhassan and E.O. Kahya, International Journal of Geometric Methods in Modern Physics **11** 1450061 (2014).
24. "Testing a Dilaton Gravity Model using Nucleosynthesis," S. Boran and E. O. Kahya, Advances in High Energy Physics, 282675 (2014).
25. "FRW cosmology with the extended Chaplygin gas," B. Pourhassan and E.O. Kahya, Advances in High Energy Physics, 231452 (2014).
26. "Extended Chaplygin gas model," B. Pourhassan and E.O. Kahya, Res. in Phys., **4** 101 (2014).
27. "The Coincidence Limit of the Graviton Propagator in de Donder Gauge on de Sitter Background," E.O. Kahya, S. P. Miao and R.P. Woodard, J. Math. Phys. **53** (2012) 022304.
28. "A useful guide for gravitational wave observers to test modified gravity models," E. O. Kahya, Phys.Lett.**B701** (2011) 291.
29. "The Zeta-Zeta Correlator Is Time Dependent," E.O. Kahya, V.K. Onemli and R.P. Woodard, Phys.Lett.**B694** (2010) 101.
30. "A Completely Regular Quantum Stress Tensor with $w < -1$," E.O. Kahya, V.K. Onemli and R.P. Woodard, Phys. Rev. **D81** (2010) 023508.
31. "A Decisive test to confirm or rule out the existence of dark matter emulators using gravitational wave observations," E. O. Kahya, Class Quantum Grav. **25** (2008) 184008.
32. "Reduced time delay for gravitational waves with dark matter emulators," S. Desai, E. O. Kahya and R. P. Woodard, Phys. Rev. **D77** (2008) 124041.
33. "Scalar Field Equations from Quantum Gravity during Inflation," E. O. Kahya and R. P. Woodard, Phys. Rev. **D77** (2008) 084012.
34. "Quantum Gravity Corrections to the One Loop Scalar Self-Mass during Inflation," E. O. Kahya and R. P. Woodard, Phys. Rev. **D76** (2007) 124005.
35. "A Generic Test of Modified Gravity Models which Emulate Dark Matter," E. O. Kahya and R. P. Woodard, Phys.Lett.**B652** (2007) 213.
36. "Quantum Instability of a $w < -1$ phase of Cosmic Acceleration," E. O. Kahya and V. K. Onemli, Phys. Rev. **D76** (2007) 043512.
37. "One Loop Corrected Mode Functions for SQED during Inflation," E. O. Kahya and R. P. Woodard, Phys. Rev. **D74** (2006) 084012.

38.“ Charged Scalar Self-Mass during Inflation,”
E. O. Kahya and R. P. Woodard, Phys. Rev. **D72** (2005) 104001.

39.“ Higher dimensional metrics of colliding gravitational plane waves,”
M. Gurses, E. O. Kahya, A. Karasu, Phys. Rev. **D66** (2002) 024029.

Journals Refereed For

Physical Review D, Physics Letters B, Classical and Quantum Gravity, and Astroparticle Physics.

Invited Talks at Workshops and Conferences

1. Hot Topics in Modern Cosmology Spontaneous Workshop XI, Corsica, France, May 1–6, 2017.
Talk: “Loop Effects on Non-Gaussianity”.
2. Utrecht Cosmology Symposium 2016, Utrecht, the Netherlands, June 27–July 1, 2016.
Talk: “Loop Corrections to Primordial Non-Gaussianities”.
3. Ankara YEF Days 2015, February 13, 2015. Talk: “Cosmological Perturbations”.
4. Feza Gürsey Summer Schools, September 1–12, 2014. Talk: “Primordial Nucleosynthesis”.
5. 12th Workshop on Quantization, Dualities and Integrable Systems, Koç University, Istanbul, Turkey, April 20–22, 2012. Talk: “Power Spectrum - Can it really be time-dependent?”.
6. Physics of de Sitter Spacetime, Albert Einstein Institute, Hannover, Germany, September 11–14, 2012.
Talk: “Quantum Effects on Scalar Particles During Inflation”.
7. IZYEYF 2012, Izmir, Turkey, June 19–23, 2012.
Talk: “Quantum Effects During Inflation”.
8. 11th Workshop on Quantization, Dualities and Integrable Systems, Pamukkale, Turkey, April 21–23, 2012.
Talk: “Quantum Effects During Inflation”.
9. Annual Meeting GRK 1523: Quantum and Gravitational Fields, Oppurg, Germany, March 23–24, 2012.
Talk: “Realizing Quantum Effects During Inflation”.
10. Quantum Gravity: From UV to IR, CERN, Geneva, Switzerland, September 5–23, 2011.
Talk: “Effects of Loops During Inflation” .
11. Modified Gravity Approaches to the Dark Sector, Strasbourg, France, June 28–July 1, 2010.
Talk: “Testing New Physics with Gravitational Waves”.
12. Gravitational Wave Tests of Alternative Theories of Gravity in the Advanced Detector Era, Milwaukee, Wisconsin, USA, June 26–27, 2010.
Talk: “Externally Triggered GW Searches and Tests of Alternate Gravity Models”.
13. Interactions in the Dark: Physics of Dark Energy–Dark Matter Interactions, Leiden, the Netherlands, April 4–9, 2009. Talk: “Testing Alternative Gravity Models”.

Invited Seminars and Conference Presentations

1. **GR21**, 21st International Conference on General Relativity and Gravitation, Columbia University, New York, USA, July 10–15, 2016.
Talk: “Effects of Time-Dependent Scalar Mode Functions on Non-Gaussianity”
2. **Planck 2016**, 19th International Conference from the Planck Scale to the Electroweak Scale, Valencia, Spain, May 23–27, 2016. Talk: “Zeta-Zeta propagator: Is it time-dependent?”
3. Pusan National University, Busan, South Korea, August 27, 2015.
Invited Talk: Zeta Zeta Correlator is Time-Dependent
4. CosKASI (Cosmology Group at KASI) Korea Astronomy and Space Science Institute, Daejeon, South Korea, August 25, 2015.
Invited Talk: Zeta Zeta Correlator is Time-Dependent
5. **Planck 2015**, 18th International Conference from the Planck Scale to the Electroweak Scale, Ioannina, Greece, May 25–29, 2015. Talk: “Zeta-Zeta propagator: Is it time-dependent?”
6. University of Granada, Granada, Spain, April 2015.
Invited Talk: “Quantum Effects During Inflation”
7. Middle East Technical University, Ankara, Turkey, November 2014.
Invited Talk: “Quantum Gravitational Effects during the Early Universe”
8. The 18th Annual International Conference on Particle Physics and Cosmology, **Cosmo 2014**, Kavli Institute for Cosmological Physics (KICP), Chicago, USA, August 2014
Talk: “Quantum Effects on Conformal Scalars during Inflation”
9. Dark Matter in the Universe and Universal Properties of Galaxies: Theory and Observations Paris, France, June 8–11, 2010.
Invited Talk: “Externally Triggered GW Searches and Tests of Alternate Gravity Models”
10. **12th** Gravitational Wave Data Analysis Workshop, MIT, Cambridge, MA, USA, December 13–16, 2007.
Talk: “Testing Modified Gravity Models by Gravitational Wave Observation”
11. **Inaugural Conference**, Penn State University, University Park, PA, USA, August 2007.
Talk: “A Generic Test of Modified Gravity Models which Emulate Dark Matter”
12. University of Crete, Heraklion, Crete, August 2006.
Invited Talk: “Quantum Corrected Mode Functions for SQED during Inflation”
13. **MG11**, Eleventh Marcel Grossmann Meeting of General Relativity, Berlin, Germany, July 2006.
Talk: “Effective Field Equations for Charged Scalar during Inflation”

Media Coverage

- Physics World, “Happy Dark Matter Day,” October 31 2017.
- Smithsonian Magazine, “What the Neutron Star Collision Means for Dark Matter,” October 31 2017.
- MIT Technology Review, “The Best of the Physics arXiv,” October 28 2017.
- Forbes, “Merging Neutron Stars Deliver Deathblow To Dark Matter And Dark Energy Alternatives,” October 25 2017.
- Ars Technica, “Colliding neutron stars apply kiss of death to theories of gravity,” October 25 2017.
- Alphr, “The neutron star merger is good news for dark matter,” October 25 2017.
- New Scientist, Particle race could settle dark matter debate, March 7, 2008.